

Preclass Learning: Base R Programming (ICA 10)

1. Base R Data Structures

- **Vector:** The fundamental R object; a sequence of elements of the *same* type.
- **Data Frame:** A list of vectors of equal length, the standard structure for tabular data.
- **List:** The most flexible structure; an ordered collection of objects, which can be of **different** types (e.g., numeric vectors, data frames, text).

2. Data Access and Subsetting

Base R primarily uses square brackets (`[]`) and the dollar sign (`$`) for accessing and modifying data.

- **Single Bracket `[]`:** Used for subsetting vectors and data frames. Returns the object type of the input (e.g., subsetting a data frame returns a data frame).
 - `df[row_index, col_index]`
- **Double Bracket `[[ ]]`:** Used primarily with lists to extract a single element. **Always** returns the content *inside* the list element, not the list element itself.
- **Dollar Sign `$`:** Used to access a column by name in a data frame or an element by name in a list (e.g., `my_data$age`).

3. Base R for Iteration and Application

Instead of `purrr::map()`, Base R uses functions in the `apply` family:

- **`lapply()`:** Applies a function to every element of a list or vector and returns a **list**.
- **`sapply()`:** Applies a function to every element and attempts to simplify the result (e.g., to a vector or array).
- **`vapply()`:** Similar to `sapply()`, but requires the user to specify the output type and length, making it safer and faster.